

Deepak Mallampati

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PROFESSIONAL SUMMARY

Applied AI / ML engineer shipping production healthcare AI end to end: HIPAA-conscious RAG (~35% retrieval precision, >90% grounded alignment), agentic LLM workflows w/ schema-validated JSON contracts (>30% hallucination reduction), predictive ML pipelines (15-20% recall on high-risk patient categories at >90% precision), and FastAPI + Docker packaging (~30% post-deploy defect reduction). Production computer vision via YOLOv8 (~30% inference latency reduction). Master's in Computer Science, Kent State. Targeting Metriport's Founding AI Engineer seat - production RAG over clinical knowledge w/ recursive semantic chunking + transformer embeddings (~35% retrieval precision, >90% grounded alignment), structured data extraction discipline transferable from EHR extracts + appointment + ticket logs (dataset reliability >98%, downstream instability -40%), Agentic Healthcare Claims 5-stage pipeline w/ schema-validated JSON contracts between agents (direct mirror to Metriport's structured-extraction problem), HIPAA-conscious governance (de-id, lineage docs, evaluation audit trails), and FastAPI/Docker packaging translate 1:1 to Metriport's open-source healthcare data interoperability platform. Ready to ramp on FHIR/TEFCA. Open to SF/Bay Area relocation; F-1 OPT (sponsorship welcome per posting).

CORE COMPETENCIES

Healthcare Data Intelligence • Structured Extraction (HTML/PDF/Text) • HIPAA-conscious Governance • Production RAG / Grounding • EHR Preprocessing + Lineage Docs • Agentic LLM Workflows • Python / FastAPI / Docker • Open-Source Mindset • Multi-Format Data Pipelines • F-1 OPT (Sponsorship Welcome)

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Applied AI, Healthcare

USA - Healthcare Technology, Applied AI

- Developed **agentic LLM workflows** for multi-step healthcare queries (eligibility checks, care workflow navigation, documentation clarification) with structured reasoning, tool discipline, grounding rules; agent response stability ~25%; hallucinations >30%.
- Built **Retrieval-Augmented Generation (RAG)** for clinical knowledge retrieval and healthcare documentation search; recursive semantic chunking + transformer embeddings improved retrieval precision ~35%; retrieval-grounded response alignment >90%.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show, care engagement scoring, support prioritization; recall +15-20% on high-risk categories, precision held >90% via class weighting, stratified sampling, threshold calibration.
- Packaged ML/LLM inference as **FastAPI / Flask** services on **Docker** with structured logging and load simulation; post-deploy defects ~30%.
- EHR + appointment + ticket preprocessing (Pandas, NumPy); dataset reliability >98%, downstream instability -40%.
- HIPAA-conscious governance**: de-identification, data lineage, evaluation audit trails, stakeholder-facing system-limitation docs.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

Hyderabad, India - Oil & Gas, Enterprise ERP

- Optimized SQL and **T-SQL stored procedures** on the Synthesis Order-to-Cash platform; query execution time ~20%, data retrieval latency ~25%.
- Built **CI/CD pipelines with Jenkins** for schema updates and stored procedure deployments; deployment errors >30%, release cycle ~35-40%.
- Designed **SQL Server DMVs and Grafana** performance dashboards for long-running queries, deadlocks, CPU/IO contention; incident recurrence ~25%, RCA speed ~30%.
- Tuned database maintenance (index rebuilds, partition-aware indexing); reconciliation runtime ~15%, deadlocks ~15%.

Machine Learning Engineer

Hyderabad, India - Nonprofit, Video Analytics, Applied NLP

- Built **transformer-based video summarization** over 5,000+ recorded sessions; hierarchical summarization + timestamp-aligned clip extraction cut manual review **60-70%**.
- AI-selected highlights aligned with human-curated moments **~85%**.
- Implemented **document Q&A with RAG**; hallucinations **~30%**, contextual accuracy **>85%**.
- Deployed as **Flask** API with a lightweight web interface for non-technical staff (research-to-production handoff).

PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Five-stage multi-agent pipeline (Intake Normalization, Validation, Consistency Analysis, Duplicate Detection, Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policies. Audit-ready explainable risk scoring with reasoning traces.

Python, LangChain, FastAPI, pgvector, GPT-4, schema contracts, vector search

Manga Lens - Multi-Provider AI Vision Chrome Extension

Real-time manga translation extension shipped to the Chrome Web Store. Multi-provider abstraction (4 AI vision providers - Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v) with per-provider payload handling for failure isolation. Manifest V3 service workers; 7-day translation cache; cost-aware payload routing (WebP for cloud, JPEG for local).

TypeScript, Manifest V3, Service Workers, OpenAI / Anthropic / Google / Ollama

Dream Decoder - Multimodal Creative Intelligence Platform

Full-stack **FastAPI + React/TypeScript/Vite/Tailwind** app coordinating multimodal AI APIs (Perplexity Sonar interpretation + Replicate image generation). Intermediate **structured prompt transformation layers** raised contextual alignment **~30%** and first-pass image success **~25-30%** over naive direct-prompt orchestration.

FastAPI, React, TypeScript, Perplexity Sonar, Replicate, multimodal orchestration

Driver Drowsiness Detection - Real-Time YOLOv8 Fatigue Monitoring

Replaced two-stage CNN pipeline with a unified **YOLOv8** detection-and-classification model - inference latency **~30%**. Sliding-window confidence aggregation suppressed blink-driven false positives **~25%**; adaptive frame skipping + NMS tuning for stable real-time operation via OpenCV video capture.

PyTorch, YOLOv8, OpenCV, LabelImg, real-time inference

EDUCATION

Master of Science, Computer Science - Kent State University

Aug 2023 - May 2025

Focus: Applied Machine Learning, NLP, Database Systems.

Bachelor of Technology, Computer Science - KL University

Jun 2019 - May 2023

Founder / Lead of E-Farming platform (university project).

CERTIFICATIONS

Deep Learning Specialization - *DeepLearning.AI* (Coursera)

2024

SKILLS

Languages: Python, TypeScript, JavaScript, SQL (T-SQL, PostgreSQL, SQLite), C++

LLM / GenAI: RAG, agentic workflows, structured outputs, evaluation pipelines, embeddings, vector search, LangChain, LlamaIndex, multi-provider AI integration, schema-validated JSON agent contracts

ML Frameworks: PyTorch, scikit-learn, XGBoost, Hugging Face Transformers, Diffusers, threshold calibration, walk-forward validation

Computer Vision: YOLOv8, OpenCV, ControlNet, OpenPose / MediaPipe, Stable Diffusion, video analytics

Backend / Inference: FastAPI, Flask, REST, Docker, structured logging, load simulation

Data: PostgreSQL + pgvector, Pandas, NumPy, EHR preprocessing, time-series

DevOps: Docker, Jenkins, GitHub Actions, Grafana, cloud-ready deployment

Domains: Applied AI, Healthcare AI, HIPAA-conscious systems, Agentic / Multi-Step LLM, Production RAG, Multimodal Systems